

Lab Sheet 05

IT23165120

Exercise 01

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     var_company stock.company%TYPE := 'IBM';
  3     var_price   stock.price%TYPE;
  4 BEGIN
  5     SELECT price
  6     INTO var_price
  7     FROM stock
  8     WHERE company = var_company;
  9
 10     DBMS_OUTPUT.PUT_LINE('Company: ' || var_company);
 11     DBMS_OUTPUT.PUT_LINE('Current Price: ' || var_price);
 12 END;
 13 /
Company: IBM
Current Price: 70

PL/SQL procedure successfully completed.
```

Exercise 02

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2   var_company stock.company%TYPE := 'IBM';
  3   var_price   stock.price%TYPE;
  4 BEGIN
  5   SELECT price INTO var_price
  6   FROM stock
  7   WHERE company = var_company;
  8
  9   IF var_price < 45 THEN
10     DBMS_OUTPUT.PUT_LINE('Current price is very low !');
11
12   ELSIF var_price < 55 THEN
13     DBMS_OUTPUT.PUT_LINE('Current price is low !');
14
15   ELSIF var_price < 65 THEN
16     DBMS_OUTPUT.PUT_LINE('Current price is medium !');
17
18   ELSIF var_price < 75 THEN
19     DBMS_OUTPUT.PUT_LINE('Current price is medium high !');
20
21   ELSE
22     DBMS_OUTPUT.PUT_LINE('Current price is high !');
23   END IF;
24 END;
25 /
Current price is medium high !
```

Exercise 03

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2   i NUMBER := 9;
  3 BEGIN
  4   LOOP
  5     DBMS_OUTPUT.PUT_LINE(RPAD(i, i, i));
  6     i := i - 1;
  7     EXIT WHEN i = 0;
  8   END LOOP;
  9 END;
10 /
999999999
888888888
77777777
6666666
55555
4444
333
22
1
```

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2   i NUMBER := 9;
  3 BEGIN
  4   WHILE i > 0 LOOP
  5     DBMS_OUTPUT.PUT_LINE(RPAD(i, i, i));
  6     i := i - 1;
  7   END LOOP;
  8 END;
  9 /
999999999
888888888
77777777
6666666
55555
4444
333
22
1
```

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> BEGIN
  2     FOR i IN REVERSE 1..9 LOOP
  3         DBMS_OUTPUT.PUT_LINE(RPAD(i, i, i));
  4     END LOOP;
  5 END;
  6 /
999999999
888888888
7777777
6666666
55555
44444
333
22
1

```

Exercise 04

```

SQL> SET SERVEROUTPUT ON;
SQL>
SQL> BEGIN
  2     FOR rec IN (SELECT clno, company, pdate, qty FROM purchase)
  3     LOOP
  4         IF rec.pdate < DATE '2000-01-01' THEN
  5             UPDATE purchase
  6             SET qty = qty + 150
  7             WHERE clno = rec.clno AND company = rec.company AND pdate = rec.pdate;
  8
  9         ELSIF rec.pdate < DATE '2001-01-01' THEN
 10             UPDATE purchase
 11             SET qty = qty + 100
 12             WHERE clno = rec.clno AND company = rec.company AND pdate = rec.pdate;
 13
 14         ELSIF rec.pdate < DATE '2002-01-01' THEN
 15             UPDATE purchase
 16             SET qty = qty + 50
 17             WHERE clno = rec.clno AND company = rec.company AND pdate = rec.pdate;
 18         END IF;
 19     END LOOP;
 20
 21     DBMS_OUTPUT.PUT_LINE('Bonus update completed. ');
 22 END;
 23 /
Bonus update completed.

```

Exercise 05

```
SQL> SET SERVEROUTPUT ON;
SQL>
SQL> DECLARE
  2     CURSOR pur_cur IS
  3         SELECT clno, company, pdate, qty FROM purchase;
  4
  5     rec pur_cur%ROWTYPE;
  6 BEGIN
  7     OPEN pur_cur;
  8
  9     FETCH pur_cur INTO rec;
 10
 11     WHILE pur_cur%FOUND LOOP
 12
 13         IF rec.pdate < DATE '2000-01-01' THEN
 14             UPDATE purchase
 15             SET qty = qty + 150
 16             WHERE clno = rec.clno AND company = rec.company AND pdate = rec.pdate;
 17
 18         ELSIF rec.pdate < DATE '2001-01-01' THEN
 19             UPDATE purchase
 20             SET qty = qty + 100
 21             WHERE clno = rec.clno AND company = rec.company AND pdate = rec.pdate;
 22
 23         ELSIF rec.pdate < DATE '2002-01-01' THEN
 24             UPDATE purchase
 25             SET qty = qty + 50
 26             WHERE clno = rec.clno AND company = rec.company AND pdate = rec.pdate;
 27         END IF;
 28
 29         FETCH pur_cur INTO rec;
 30     END LOOP;
 31
 32     CLOSE pur_cur;
 33
 34     DBMS_OUTPUT.PUT_LINE('Bonus update completed using explicit cursor.');
```

Bonus update completed using explicit cursor.

Start